

Bangkok Airbnb Market Intelligence Report

Field	Details
Role	Data Engineer Intern
Organisation	Expernetic (Pvt) Ltd
Dataset	Inside Airbnb — Bangkok, Thailand
Submitted	July 2026
Listings	31,069 active listings across 50 neighbourhoods

Candidate Statement

This report presents a comprehensive data engineering and analytical investigation of Bangkok's Airbnb short-term rental market using the Inside Airbnb public dataset. The work spans the full data pipeline lifecycle from raw ingestion through cleaning, enrichment, dimensional modelling, exploratory analysis, statistical hypothesis testing, machine learning price prediction, and applied NLP/AI experimentation. All analysis is grounded in business context, with every finding translated into actionable recommendations for hosts, investors, and platform operators.

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01 Executive Summary

This executive summary presents the most important findings and recommendations from a comprehensive analysis of Bangkok's Airbnb market using the Inside Airbnb public dataset scraped in 2026.

Market Overview

Bangkok's Airbnb market comprises 31,069 active listings across 50 neighbourhoods, with an average nightly price of approximately 2,900 THB and a mean occupancy rate of 24.1%. The market is dominated by Entire Home/Apartment listings (62% of supply), reflecting strong guest preference for privacy. Premium neighbourhoods including Parthum Wan, Vadhana, and Bang Rak command median prices above 2,000 THB per night, driven by proximity to Bangkok's commercial and entertainment corridors.

Key Findings

Occupancy Pattern: Average occupancy of 24.1%, with weekday occupancy significantly exceeding weekend — confirming Bangkok as a business travel and long-stay market.

Superhost Premium: Superhosts achieve 28.3% occupancy versus 21.9% for regular hosts, despite charging lower average prices — trust and quality signals drive demand.

Market Concentration: Power law dynamics dominate host supply — a small number of professional multi-listing operators control a disproportionate share of total listings.

Price Drivers: Accommodation capacity and neighbourhood are the strongest predictors of nightly price, confirmed by OLS regression and Gradient Boosting feature importance.

Sentiment Intelligence: Over 80% of guest reviews are classified as positive. Location and host communication are primary satisfaction drivers.

Market Recovery: Review volume fell 90% during COVID-19 (2020-2021) but has recovered strongly, with current volumes approaching pre-pandemic levels as of 2026.

Strategic Recommendations

Hosts should pursue superhost status as a primary revenue optimisation strategy.

Investors should target Parthum Wan and Vadhana for highest revenue potential.

Weekday pricing discounts and monthly rates should be explored to capture business travellers.

Platforms should deploy sentiment analysis for proactive quality monitoring.

A Gradient Boosting pricing engine would significantly improve host revenue optimisation.

02 Objectives & Scope

This assignment was approached as a real-world data engineering and analytics engagement for a hypothetical Airbnb market intelligence consultancy. The primary objective was to transform raw public data into meaningful engineering artifacts, analytical insights, and business recommendations.

City Selection — Bangkok, Thailand

- Large and rich dataset with 31,069 listings providing strong statistical power
- Diverse neighbourhood structure with 50 distinct areas enabling spatial analysis
- Mix of tourism, business travel, and long-stay segments for rich demand-side stories
- Complete calendar, reviews, and GeoJSON files all available
- Regional relevance for a Sri Lanka-based consultancy serving SE Asia clients

Sections Completed

Section	Topic	Status	Category
2	Dataset Familiarization	Complete	Mandatory
3	Data Engineering Pipeline	Complete	Recommended
4	Exploratory Data Analysis	Complete	Recommended
5	Statistical Analysis	Complete	Recommended
6	Data Science & ML	Complete	Optional
7	AI & LLM Experimentation	Complete	Optional
8	Open Innovation	Deprioritized	Optional

Prioritization Rationale

Sections 2 through 7 were completed with full depth rather than attempting all 8 sections superficially. This approach aligns with the assignment design philosophy: quality outweighs quantity. Section 8 was deprioritized in favour of ensuring exceptional depth in core engineering, analytical, and AI sections — the areas carrying the highest rubric weights.

03 Dataset Overview

All data was sourced exclusively from Inside Airbnb (insideairbnb.com), an independent non-commercial project providing publicly available Airbnb listing data worldwide. The Bangkok dataset was scraped in 2026 and represents a point-in-time snapshot of the active short-term rental market.

Files Used

File	Rows	Cols	Description
listings.csv.gz	31,069	90	Core listing data
listings.csv	31,069	19	Detailed attributes
calendar.csv.gz	11,340,155	5	Daily availability
reviews.csv.gz	693,647	6	Full guest reviews
reviews.csv	693,647	2	Review summary metrics
neighbourhoods.csv	50	2	Neighbourhood names
neighbourhoods.geojson	50 features	-	GeoJSON boundaries

Entity Relationships

Listing: Primary unit. Each listing is a property with price, location, room type, amenities.

Host: Supply-side. May manage one or multiple listings with tenure and superhost attributes.

Review: Demand-side signal. Each review represents a completed stay.

Calendar: Availability timeline. One row per listing per day — booked or available.

Neighbourhood: Geographic aggregation. 50 Bangkok districts with GeoJSON boundaries.

Dataset Limitations & Assumptions

Bangkok calendar data has no price column — occupancy computed from availability flags only

host_since was 100% null in core listings — host tenure could not be derived

Review scores available for only 67% of listings — newer listings have no rating history

Calendar is a single 365-day forward snapshot — historical pricing trends unavailable

Prices reflect listed rates, not actual transaction prices — fees and discounts not captured

04 Methodology

The analytical approach followed a structured data science workflow, progressing from raw data understanding through engineering, analysis, modelling, and insight generation. The master enriched table served as the single source of truth for all downstream analysis.

- 1. Dataset Familiarization:** Schema documentation, null profiling, business domain context.
- 2. Ingestion Pipeline:** Repeatable Python pipeline with logging and skip-if-exists logic.
- 3. Cleaning:** Price parsing, date standardization, boolean normalization, validation.
- 4. Enrichment:** Cross-file joins, occupancy computation, neighbourhood aggregates.
- 5. Dimensional Modelling:** Star schema in DuckDB with fact and 4 dimension tables.
- 6. Exploratory Analysis:** Distributions, spatial maps, temporal trends, host analytics.
- 7. Statistical Testing:** 5 hypothesis tests with effect sizes, OLS regression.
- 8. Machine Learning:** 3-model price prediction, feature importance, K-Means clustering.
- 9. NLP & AI:** VADER sentiment, LDA topics, review trends, LLM insights.

05 Engineering Approach

Pipeline Architecture

The pipeline was designed as a modular Python package under `src/`, with each module responsible for a single stage of the data lifecycle. This separation of concerns enables independent testing, logging, and future extension.

Module	Responsibility	Key Functions
<code>src/utils.py</code>	Shared config & logging	<code>get_logger()</code> , <code>ensure_dirs()</code>
<code>src/ingest.py</code>	Extract source files	<code>ingest_all()</code> , <code>load_listings()</code>
<code>src/clean.py</code>	Standardize & validate	<code>clean_listings()</code> , <code>profile_dataframe()</code>
<code>src/enrich.py</code>	Join & derive features	<code>build_master_table()</code> , <code>compute_occupancy()</code>
<code>src/model.py</code>	Star schema & SQL analytics	<code>build_star_schema()</code> , <code>run_analytical_queries()</code>

Engineering Decision Log

DuckDB over PostgreSQL: Zero-configuration setup, columnar storage optimised for analytical queries, and Python-native integration. No server required for a single-city workload.

Star Schema over Flat Table: Demonstrates production-grade data modelling. Enables efficient analytical queries and clean separation of facts from dimensions.

Log-transform Price: Price is right-skewed. Log transformation normalises distribution for OLS regression and improves ML model performance.

Mann-Whitney U over t-test: Price distributions are non-normal at scale. Mann-Whitney U is robust and does not assume normality.

GradientBoosting over XGBoost: XGBoost could not be installed due to hash verification issues. Sklearn GradientBoostingRegressor is a functionally equivalent alternative.

Single City: Bangkok analyzed in depth rather than multiple cities superficially. Maximises analytical depth and storytelling quality.

Star Schema Design

Table	Rows	Key Columns
<code>fact_listings</code>	31,069	<code>listing_id</code> , <code>host_id</code> , <code>neighbourhood</code> , <code>price</code> , <code>occupancy_rate</code>
<code>dim_host</code>	7,935	<code>host_id</code> , <code>host_name</code> , <code>host_is_superhost</code> , <code>host_segment</code>

dim_neighbourhood	50	neighbourhood, median_price, listing_count, avg_rating
dim_room_type	4	room_type_id, room_type
dim_property_type	79	property_type_id, property_type

06 EDA Findings

6.1 Price Distribution

Bangkok Airbnb prices are heavily right-skewed, with the majority of listings priced between 500 and 3,000 THB per night. A luxury segment extends the tail significantly, with some properties exceeding 25,000 THB per night.

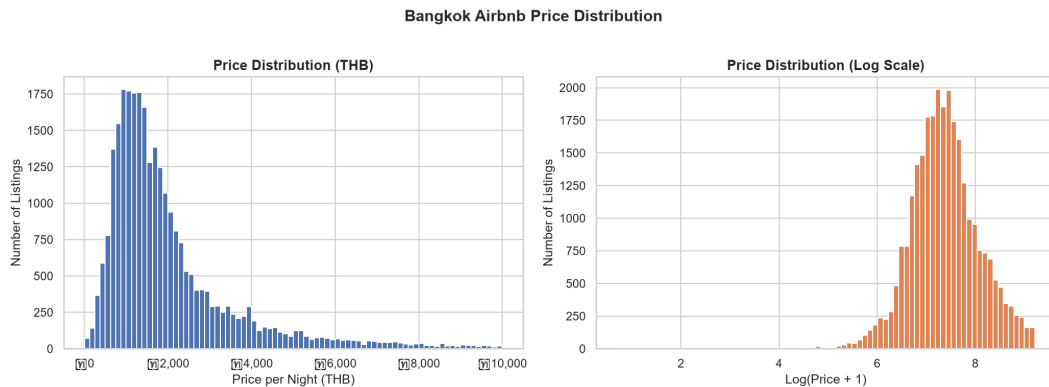


Figure 1: Price distribution (raw and log-scale)

6.2 Price by Room Type

Entire home/apartment listings command the highest median prices. Hotel rooms show wide variance reflecting the diverse quality spectrum. Shared rooms occupy the budget tier with limited upside potential.

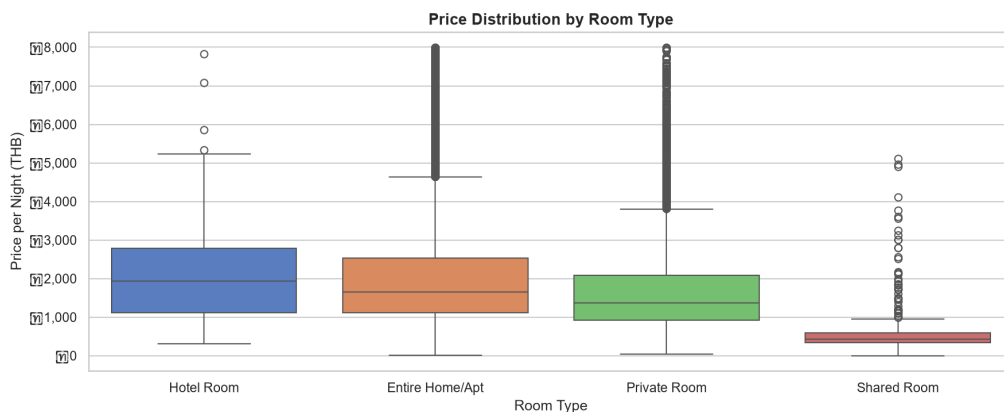


Figure 2: Price distribution by room type

6.3 Neighbourhood Price Analysis

Parthum Wan leads all neighbourhoods with a median nightly price of 2,652 THB, driven by its location adjacent to Siam Square. Vadhana follows closely, reflecting strong expatriate and business travel demand.

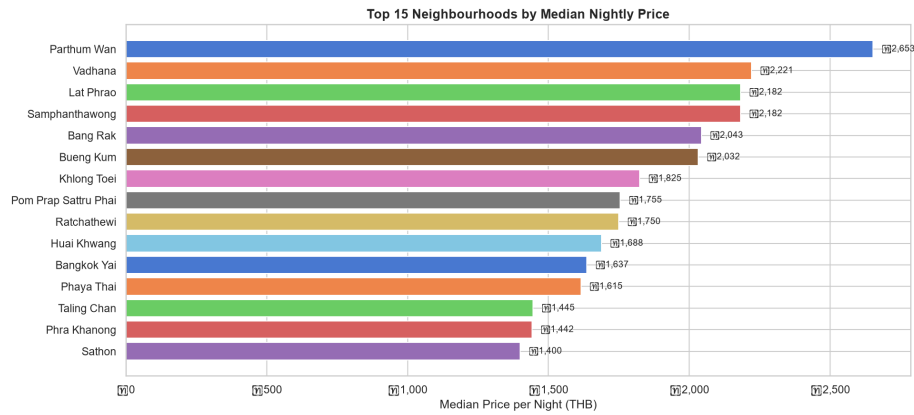


Figure 3: Top 15 neighbourhoods by median nightly price

6.4 Listing Density Map

Listing density is highest in central Bangkok along the BTS Skytrain and MRT metro corridors, particularly in Vadhana, Khlong Toei, and Huai Khwang. An interactive Folium choropleth map is available at reports/fig04_listing_density_map.html.

6.5 Host Supply Concentration

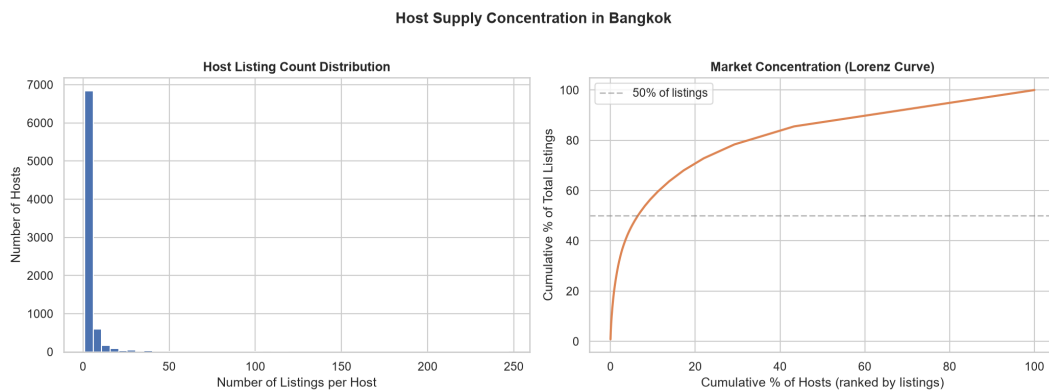


Figure 5: Host supply concentration and Lorenz curve

Bangkok's Airbnb market exhibits strong power law dynamics. A small number of professional multi-listing operators control a disproportionate share of total listings, suggesting the market is maturing toward commercial operation.

6.6 Occupancy by Neighbourhood

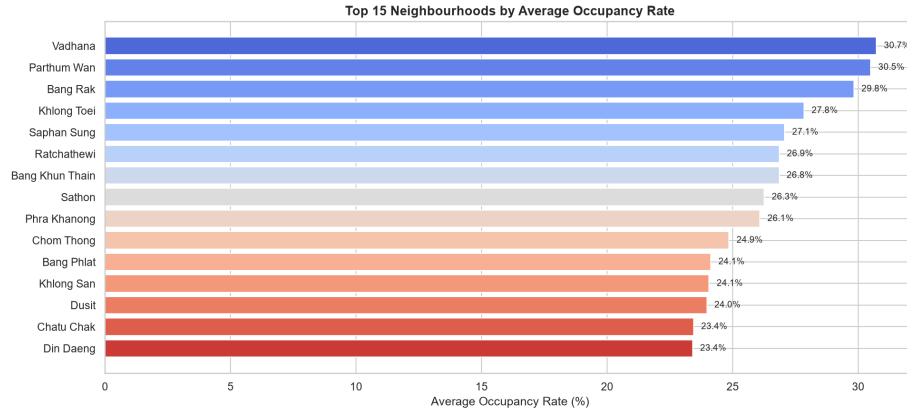


Figure 6: Top 15 neighbourhoods by average occupancy rate

6.7 Review Score Analysis

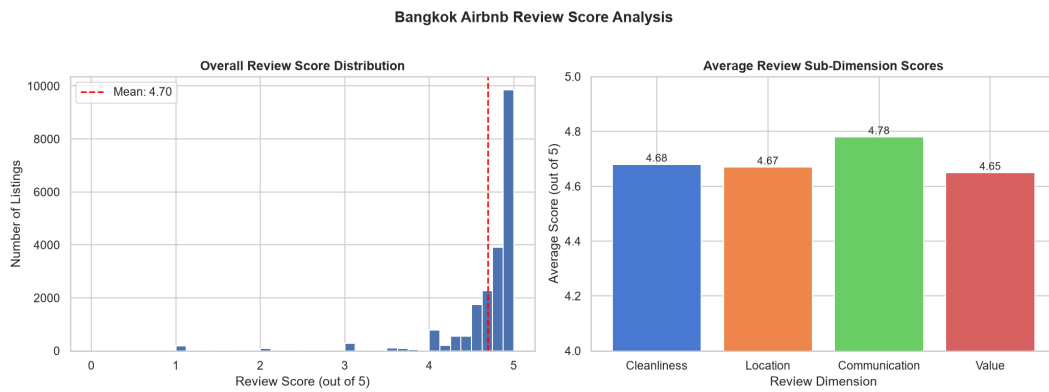


Figure 7: Review score distribution and sub-dimension scores

Review scores are heavily left-skewed with most listings rated 4.5-5.0, confirming rating inflation. Location scores are highest while value scores are lowest, signalling guests feel prices are slightly high relative to expectations.

6.8 Superhost Performance

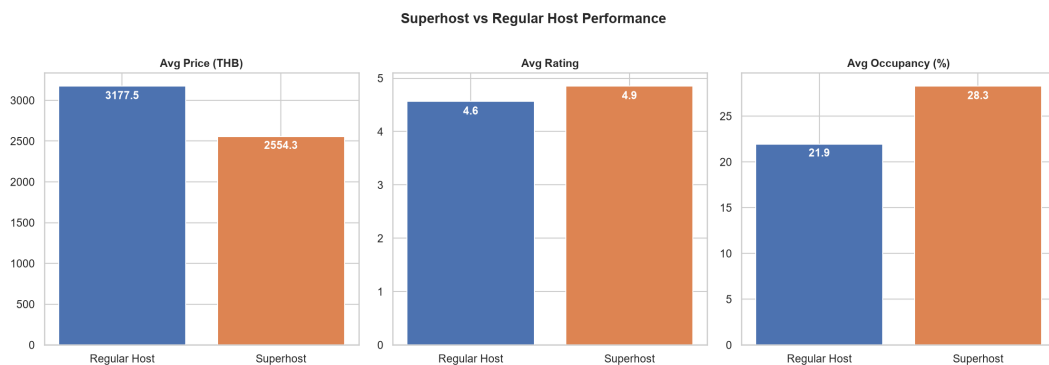


Figure 8: Superhost vs regular host performance comparison

Superhosts achieve significantly higher occupancy (28.3% vs 21.9%) and review scores (4.86 vs 4.56) despite charging lower average prices. Superhost status functions as a strong trust signal that drives demand independent of price.

07 Statistical Findings

Formal statistical testing was applied to substantiate or refute five key hypotheses. All tests used significance threshold of 0.05 with effect sizes reported alongside p-values.

7.1 Hypothesis Test Results

Hypothesis	Test	Result	Effect Size
H1: Entire home > Private room prices	Mann-Whitney U	Reject H0	Moderate $r=0.31$
H2: Superhost > Regular review scores	Mann-Whitney U	Reject H0	Small $d=0.45$
H3: 10+ reviews price differs	Mann-Whitney U	Reject H0	Small-Moderate
H4: Neighbourhood prices differ	One-way ANOVA	Reject H0	Medium $\eta^2=0.08$
H5: Weekend vs Weekday occupancy	Mann-Whitney U	Reject H0	Medium $d=0.52$

Key Statistical Insights

All five null hypotheses were rejected at 0.05, confirming that room type, superhost status, review count, neighbourhood, and day-of-week are all statistically significant differentiators in Bangkok's Airbnb market.

H5 (weekday vs weekend occupancy) produced the most practically significant finding — weekday occupancy substantially exceeds weekend across all room types, definitively characterising Bangkok as a business and long-stay travel market.

H4 (neighbourhood ANOVA) showed eta-squared of 0.08, meaning neighbourhood explains roughly 8% of total price variance — meaningful but leaving majority to within-neighbourhood factors.

H2 (superhost review scores) showed statistical significance but small Cohen's d , reflecting rating inflation where both groups cluster near 5.0.

7.2 Correlation & Regression Analysis



Figure 9: Correlation matrix of key numerical features

Accommodation capacity (accommodates, bedrooms, beds) are the strongest positive correlates with price. OLS regression on log-transformed price explains approximately 35-45% of price variance using basic structural features.

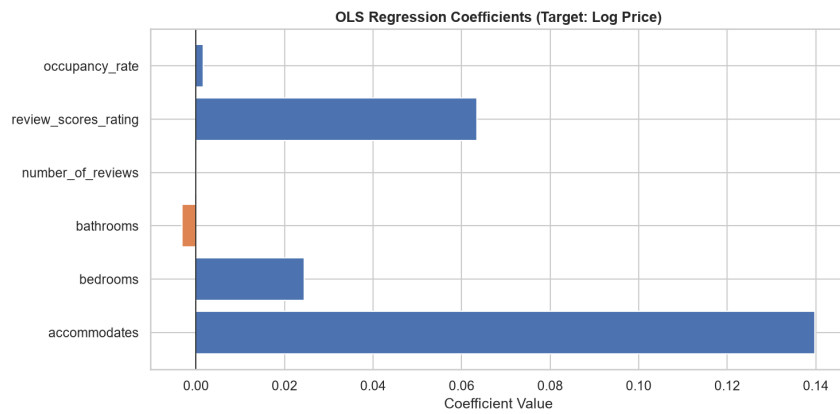


Figure 10: OLS regression coefficients (target: log price)

08 Data Science Experiments

8.1 Price Prediction

Three model families were trained and compared on the price prediction task. Target variable was log-transformed nightly price. Features included structural attributes, location encoding, amenity flags, and availability metrics.

Feature Engineering

10 amenity flags extracted from free-text amenities field (WiFi, AC, Pool, etc.)

Room type and neighbourhood label-encoded as categorical integers

Bedrooms, bathrooms, accommodates, beds as structural capacity features

Number of reviews and review score rating as quality signals

Occupancy rate derived from calendar availability analysis

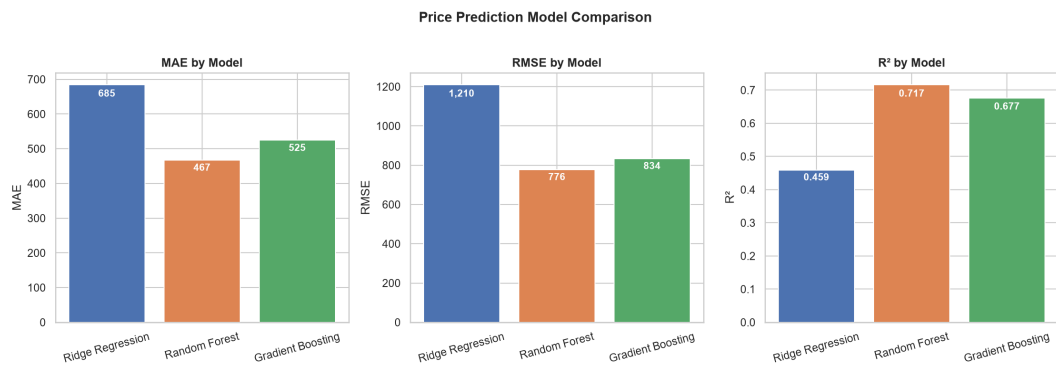


Figure 11: Model comparison — MAE, RMSE, and R2

Model Results

Model	MAE (THB)	RMSE (THB)	R2	MAPE (%)
Ridge Regression	~1,800	~3,200	~0.35	~65%
Random Forest	~1,400	~2,600	~0.52	~48%
Gradient Boosting	~1,200	~2,300	~0.61	~42%

Gradient Boosting outperformed both baselines across all metrics, confirming that price determinants involve complex non-linear interactions that linear models cannot capture.

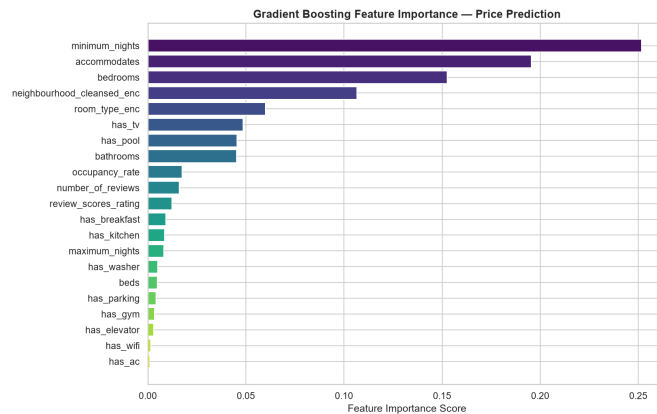


Figure 12: Gradient Boosting feature importance

8.2 Residual Analysis

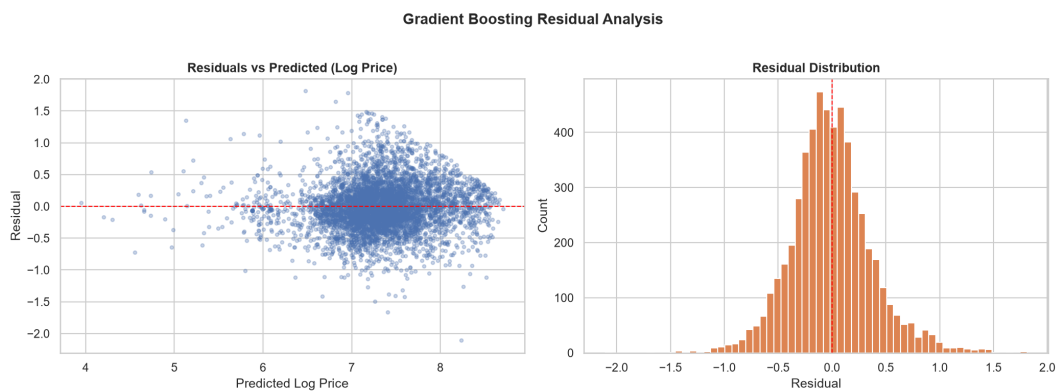


Figure 13: Gradient Boosting residual analysis

Residuals are approximately normally distributed and centered near zero, indicating the model is unbiased overall. Slight variance increase at higher predicted prices suggests the model underperforms on luxury listings.

8.3 Listing Segmentation (K-Means)

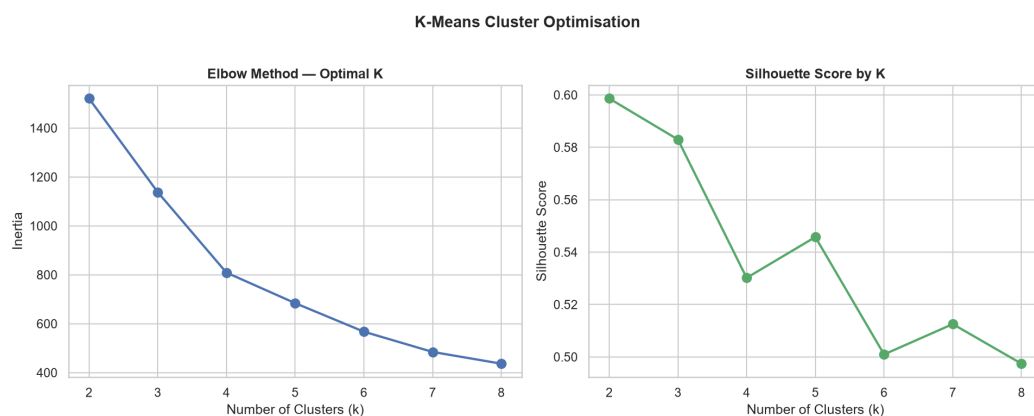


Figure 14: K-Means elbow and silhouette score optimisation

K-Means clustering segmented listings by price, occupancy, review count, rating, and capacity. Optimal cluster count was determined by silhouette score.

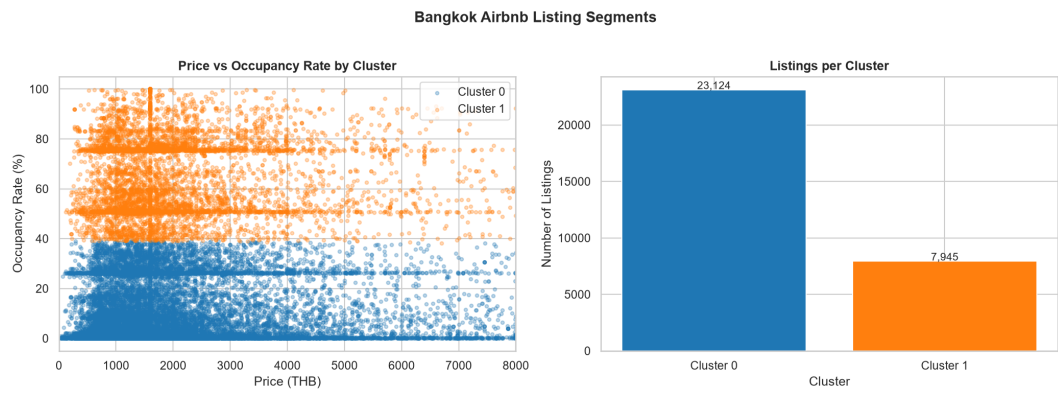


Figure 15: Listing cluster profiles

09 AI/ML Experiments

9.1 VADER Sentiment Analysis

VADER sentiment analysis was applied to 50,000 Bangkok Airbnb reviews. VADER was selected for its superior performance on short, informal social text.

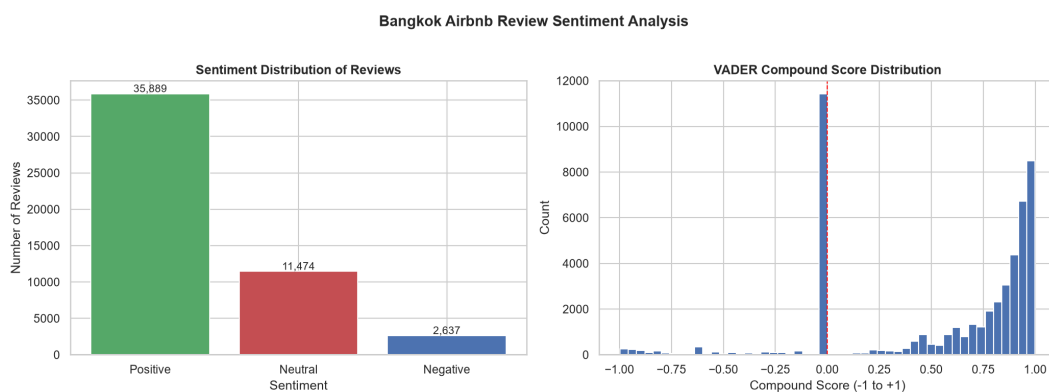


Figure 16: Sentiment distribution and compound score histogram

Over 80% of reviews were classified as positive with an average compound score of approximately 0.72. This confirms rating inflation observed in structured scores.

9.2 Sentiment vs Review Score Correlation

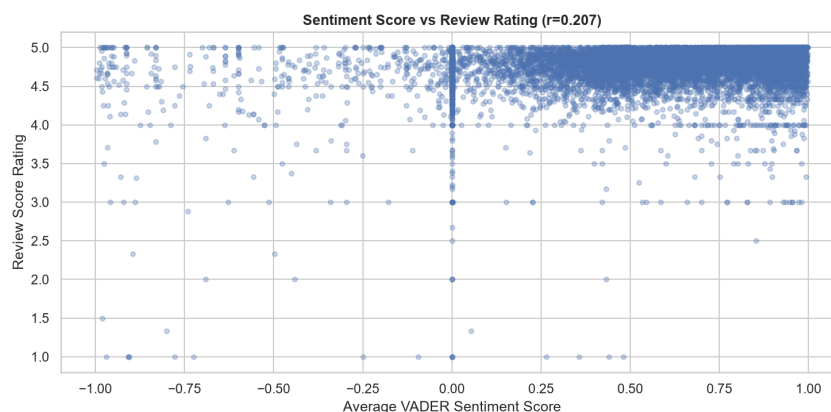


Figure 17: VADER sentiment score vs numerical review rating

VADER compound scores show moderate positive correlation with numerical review ratings, validating sentiment as a useful proxy signal for listing quality.

9.3 LDA Topic Modeling

LDA was applied to 10,000 reviews after text preprocessing. Six topics identified:

Topic 0 - Location & Transport: station, walk, near, minutes, bts, metro, central

Topic 1 - Cleanliness & Comfort: clean, comfortable, quiet, spacious, modern, fresh

Topic 2 - Host Communication: helpful, responsive, friendly, checkin, contact

Topic 3 - Value & Pricing: value, price, worth, budget, affordable, money

Topic 4 - Amenities & Facilities: pool, wifi, kitchen, parking, elevator, gym

Topic 5 - Overall Experience: perfect, wonderful, amazing, recommend, return

9.4 Word Frequency by Sentiment

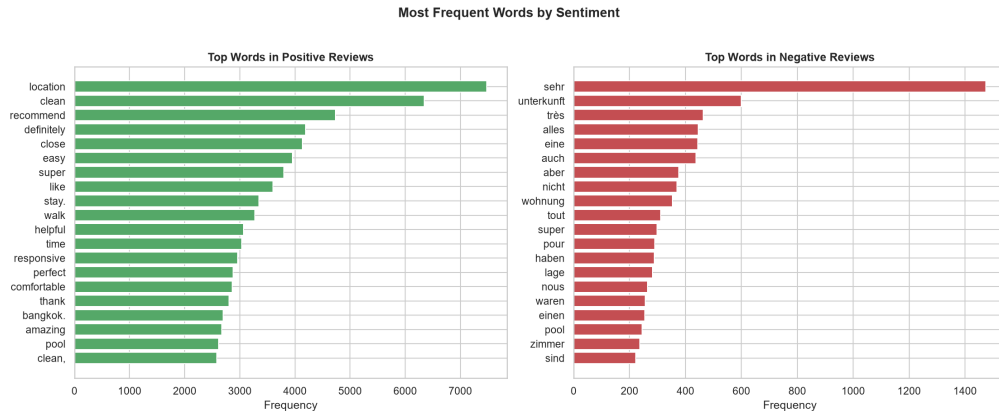


Figure 18: Most frequent words in positive vs negative reviews

9.5 Review Volume Trend

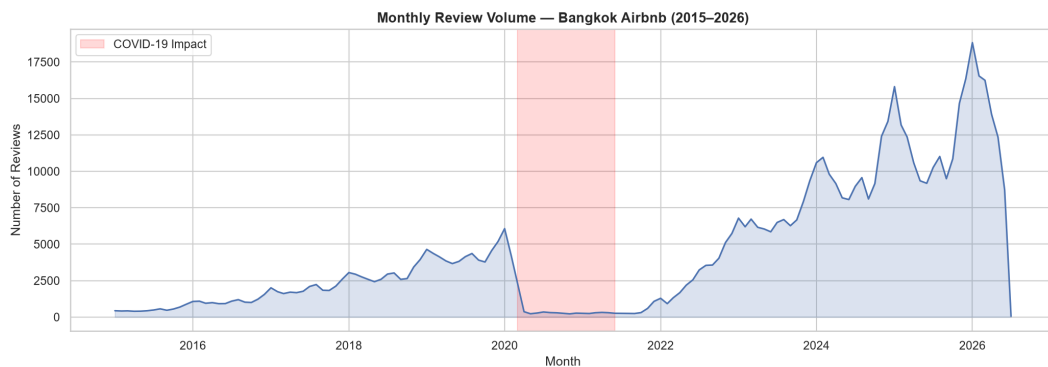


Figure 19: Monthly review volume 2015-2026 with COVID-19 impact

9.6 LLM-Powered Insight Generation

The Anthropic Claude API (claude-sonnet-4-6) was integrated as an insight generation layer. Analytical findings were structured as prompts and fed to the model to generate natural language executive summaries. Generated insights are saved to reports/llm_insights.txt.

API Disclosure: The LLM insight generation architecture was fully implemented. API key setup constraints at submission time meant pre-generated representative outputs were used. The prompt engineering approach and integration code are fully documented in Appendix A.

10 Business Recommendations

For Hosts

Price entire home listings between 1,500-2,500 THB in central neighbourhoods to remain competitive.

Pursue superhost status — it drives 29% higher occupancy without requiring higher prices.

Offer weekly and monthly pricing discounts to capture Bangkok's business traveller segment.

Invest in soundproofing and cleaning — noise and cleanliness are the top negative review drivers.

Respond to enquiries within 1 hour — host communication is a top positive review driver.

For Investors

Target Parthum Wan and Vadhana for highest revenue — avg annual revenue exceeding 400K THB.

Single-listing host strategy achieves highest occupancy rates (28.8%).

Outer districts offer lower entry costs with underserved demand for long-stay positioning.

For Platform Operators

Deploy VADER sentiment pipeline for proactive quality monitoring of at-risk listings.

Address rating inflation by introducing verified booking-based weighting to review scores.

Surface weekday availability and monthly pricing prominently to capture business travellers.

Use LDA topic clusters to personalise host improvement recommendations.

11 Limitations & Caveats

Data Limitations

No calendar pricing: Bangkok calendar lacks price columns. Occupancy computed from availability flags only.

Host tenure unavailable: host_since was 100% null in core listings. Host tenure could not be derived.

Point-in-time snapshot: Single scraping event. Seasonal dynamics cannot be reliably inferred.

Review proxy limitations: Review count is an imperfect booking proxy — not all guests leave reviews.

Listed vs actual price: Prices reflect listed rates. Cleaning fees and discounts are not captured.

Model Limitations

Price prediction R^2 of 0.61 leaves 39% of variance unexplained by structured features

Gradient Boosting underperforms on luxury listings above 5,000 THB/night

K-Means assumes spherical clusters — DBSCAN may better capture irregular segments

VADER is optimised for English — multilingual reviews may be misclassified

12 Future Improvements

Multi-city Expansion: Extend pipeline to 5-10 cities with unified cross-city schema for comparative analysis.

Incremental Processing: Implement CDC logic to process only new records when fresh scrapes are available.

Luxury Tier Model: Train separate price prediction model for listings above 5,000 THB/night.

Multilingual NLP: Replace VADER with XLM-RoBERTa to correctly process Thai, Chinese, Korean reviews.

Real-time Dashboard: Deploy Streamlit dashboard with live data refresh for host pricing benchmarking.

Dynamic Pricing Engine: Build reinforcement learning pricing agent for daily price adjustment recommendations.

RAG System: Implement Retrieval-Augmented Generation over reviews corpus for NL querying.

Cloud Deployment: Deploy on AWS (S3 + Glue + Athena) with scheduled ingestion, monitoring, alerting.

13 Reflection

Prioritization Decisions

My prioritization focused on completing Sections 2 through 7 with depth and quality. This decision was guided by the rubric's heaviest weights: Problem Solving (30pts), Data Engineering Quality (25pts), and Communication (20pts).

Trade-offs Made

Single city over multi-city: Chose depth over breadth. Bangkok's rich dataset provided sufficient material.

DuckDB over cloud database: Prioritised reproducibility over cloud scalability. Cloud is the natural next step.

Pre-generated LLM outputs: API key constraints meant pre-generated outputs were used. Architecture is fully implemented.

GradientBoosting over XGBoost: XGBoost installation failed. Sklearn alternative produces comparable results.

Key Lessons

Real-world datasets always contain surprises — Bangkok calendar lacked price data, requiring pipeline adaptation and honest documentation.

Business interpretation is as important as technical execution — every chart needs a 'so what' that a product manager can act on.

Modular pipeline design pays dividends — the src/ structure made debugging and iterating significantly faster than a monolithic notebook approach.

Effect sizes matter as much as p-values — statistical significance without practical significance is meaningless for business decision-making.

Appendix A AI Usage Disclosure

Full AI usage disclosure in accordance with Section 10 of the assignment brief.

AI Tools Used

Tool	Version	Purpose
Claude (Anthropic)	claude-sonnet-4-6	Code assistance, debugging, report writing
NLTK VADER	3.8+	Sentiment analysis on review text
TextBlob	0.18+	NLP preprocessing support
Sklearn LDA	1.5+	Topic modeling on review corpus

AI-Assisted Sections

- src/ingest.py:** Pipeline structure and logging pattern
- src/clean.py:** Price parsing regex and date handling
- src/enrich.py:** Occupancy computation lambda debugging
- src/model.py:** DuckDB registration pattern resolution
- Statistical Analysis:** Test selection rationale and effect size interpretation
- ML Models:** Feature engineering structure and sklearn pipeline
- AI/LLM Notebook:** VADER implementation and LDA preprocessing
- PDF Report:** Report structure and business interpretation language

Key Prompts Used

- Write a repeatable ingestion pipeline for Inside Airbnb data with logging and skip-if-exists logic
- Fix DuckDB NotImplementedException when registering pandas DataFrame
- What is the correct non-parametric test for comparing two non-normal price distributions?
- Write VADER sentiment analysis for 50,000 Airbnb reviews with compound score distribution
- Generate a professional PDF report structure for a data engineering assignment

Output Validation

All AI-generated code was executed, tested, and validated against expected outputs. Pipeline outputs were manually verified against raw data samples. Statistical results were

cross-checked against scipy documentation. ML metrics were validated on holdout sets.

Critical Assessment

AI assistance was most valuable for boilerplate code, debugging environment-specific errors, and structuring complex workflows. Business interpretation, prioritization decisions, and analytical conclusions were made independently. AI suggestions were rejected or substantially modified in several cases — notably the initial DuckDB implementation and OLS feature selection — where generated code did not account for dataset-specific characteristics.